

Shubham Mandora

✉ shubhammandora220800@gmail.com ☎ +1 (559) 387 7883 🔗 [linkedin.com/in/shubhamm2208](https://www.linkedin.com/in/shubhamm2208)

SUMMARY

Computer Science graduate (MS, NYIT) with experience in full-stack engineering, AI-powered system development, and cloud-native distributed systems. Skilled in Python, Django, FastAPI, Go, TypeScript, React, Docker, Kubernetes, and AWS with hands-on expertise building LLM-powered agentic workflows, RAG pipelines, and production-grade automation systems. Experienced delivering end-to-end technical solutions across distributed architectures with a strong focus on reliability, observability, and operational excellence.

PROFESSIONAL EXPERIENCE

Founding Engineer - Backend and Data

Vizzy Labs, Inc

08/2025 – Present
Brooklyn, NY

- Designed, built, and deployed large-scale backend architecture and APIs for Vizzy, a real-time mobile puzzle game on Google Play and App Store, ensuring high reliability, scalability, and SLO-backed 24x7 uptime across AWS and Azure distributed systems.
- Developed and optimized high-performance RESTful APIs for gameplay, authentication, leaderboard ranking, and financial transaction management, building automated test frameworks - unit, integration, API contract, and performance suites - to validate business logic and microservices reliability.
- Architected CI/CD pipelines with automated build verification, regression testing, and quality gates using Docker, Jenkins, GitHub Actions, and Kubernetes, improving release engineering and deployment confidence.
- Designed automated performance benchmarking frameworks and test plans for load and stress testing to measure latency, throughput, and concurrency under production-like conditions, enabling proactive bottleneck detection.
- Designed telemetry-driven observability combining SLI/SLO monitoring, Grafana dashboards, automated failure detection, and proactive alerting to support production incident response and reliability goals.

Research Assistant

New York Institute of Technology

01/2025 – 08/2025
New York City, USA

- Engineered an AI-driven real-time gesture recognition system using a custom smart glove with inductive sensors, building and deploying deep learning production pipelines in Python with model monitoring and validation.
- Integrated LLM and Generative AI capabilities into production pipelines using LangChain — including RAG-style retrieval, tool-use orchestration, and agentic workflow patterns — improving system accuracy and automating content validation during real-time inference
- Developed a Python-based data pipeline with GraphQL APIs and Dash for live session management, multi-channel data visualizations, and structured content processing via real-time data streams.
- Designed and trained ML classification models (Random Forest, SVM, MLP) in scikit-learn, performing model optimization and performance benchmarking to achieve >95% accuracy with robust validation across dataset updates.
- Developed structured technical documentation and reproducible experiment frameworks to support knowledge transfer, onboarding, and iterative model improvement across the research team.

Software Developer Intern

IDZ Digital Pvt. Ltd.

02/2022 – 04/2022
Mumbai, India

- Developed multiple consumer mobile games for the Tizi World Series using Java, Kotlin, C# and Unity, game state management, and player input handling, device state management, and sensor input handling across the full SDLC from requirements through App Store and Play Store deployment.
- Implemented dynamic front-end animations and interactive gameplay features while developing structured unit and functional test cases to support quality engineering and validate gameplay logic.
- Designed and programmed robust backend game logic to support complex game mechanics, incorporating automated validation scripts to ensure consistency, stability, and seamless integration.
- Executed systematic regression automation and performance validation prior to releases, designing test plans to identify defects early, improve product quality, and ensure deployment stability.
- Managed end-to-end integration and iterative feature releases using Agile methodologies, version control, and test-driven development practices, contributing to team knowledge sharing and technical documentation to improve release readiness.

Developer Intern

Clear Exam

05/2021 – 07/2021
Nashik, India

- Developed a high-throughput Java and Python-based data processing pipeline with automated classification workflows, applying structured testing and validation to ensure production readiness and data quality across diverse datasets.
- Designed secure API-driven backend services with authentication, input validation, and data integrity checks, ensuring consistent performance and reliability across user-facing application features.

Software Developer Intern

Divspace, LLP

07/2020 – 10/2020
Nasik, India

- Developed Python and JavaScript automation scripts to streamline property listing data processing, content validation, and API integration across the real estate web platform.
- Built CI/CD-style deployment workflows with structured data integrity checks and automated validation, ensuring consistent delivery and production readiness across iterative feature releases.

EDUCATION

Master's of Science in Computer Science

New York Institute of Technology

09/2023 – 05/2025
Manhattan, NY

Bachelor of Technology in Computer Science

Sandip University

07/2019 – 05/2022
Nasik, India

PROJECTS

Sustainable City Management

- Built a large-scale real-time data engineering and analytics platform for Dublin city infrastructure, integrating live APIs (DublinBikes, Sonitus) with automated data validation, pipeline testing, and structured unit/integration tests to ensure data integrity across high-volume data streams.
- Built a scalable backend using Django and PostgreSQL, enabling real-time data ingestion, processing, and visualization across high-volume data streams, supported by structured unit and integration testing to validate pipeline reliability and business logic.
- Deployed containerized microservices using Docker and Kubernetes, incorporating automated health checks and regression testing workflows to ensure high availability, fault tolerance, and consistent service performance.
- Automated CI/CD pipelines with Jenkins, integrating automated build verification, test execution, and quality gates to streamline deployments and reduce release risks.
- Designed a user-focused frontend using React and Django templates, executing functional and performance testing to ensure UI responsiveness, cross-browser compatibility, and accurate real-time data representation.

Tool for Machine Learning-Driven Urban Transport Forecasting

- Developed and implemented a GBDT-based prediction system in the SUMO environment, processing real-time data streams incorporating traveler preferences and environmental factors, building scalable forecasting pipelines for accurate transport mode prediction and decision-making.
- Built an end-to-end ML pipeline using PySpark and scikit-learn, performing hyperparameter tuning (GridSearchCV), model monitoring, and automated regression testing to validate model performance across dataset updates.
- Cleaned, processed, and analyzed large datasets to build predictive traffic simulation models using Multinomial Logistic Regression, Random Forest, SVM, and XGBoost, conducting systematic performance benchmarking and validation to ensure reliable, production-ready forecasting outputs for urban planning.

Online Proctoring Using AI

- Engineered a real-time agentic AI inference system using PyTorch, TensorFlow, and scikit-learn for secure online proctoring, implementing RAG-based retrieval patterns and LLM-driven anomaly detection to optimise model performance across large-scale data pipelines.
- Reviewed and optimized ML pipeline code for inference efficiency and maintainability, conducting systematic performance benchmarking to ensure scalability and adherence to best practices across production workloads.
- Developed content moderation pipelines integrating SLA-based real-time inference processing and BI-driven analytics to improve system resilience, compliance, and decision-making accuracy.

SKILLS

Programming Languages: Python, C, C#, C++, GO, TypeScript, Java, Kotlin, JavaScript, R, Swift, Shell, MATLAB

Data Engineering: PostgreSQL, MongoDB, Redis, Apache Kafka, JSON, XML, Hadoop, Neo4j, ETL, Pandas

Cloud & DevOps: AWS (EC2, S3, SageMaker, Lambda), Docker, Kubernetes, Jenkins, CI/CD, API Deployment, Azure, GCP

Visualization & BI: Power BI, Tableau, Looker

Frameworks & Web: Node.js, React, Django, FastAPI, Spring WebFlux, LangChain, GraphQL, REST APIs, HTML/CSS

Tools: Git, GitHub, Grafana, RedHat Linux, pytest, Microsoft 365

CERTIFICATES

Google Advanced Data Analytics
Coursera

Python Essentials Course badge
Cisco

AWS certified Cloud Practitioner
AWS